AirGuard: Smart IoT-Based Environmental & Confined Space Air Safety System

GUIDED BY:
MS.GRACE
ASST PROFESSOR
CSE DEPARTMENT
NCERC PAMPADY

PRESENTED BY:
MUHAMMED ANAS A(NCE22CS079)
SEKKEENA KHAN PM(NCE22CS100)
SHAHANA FATHIMA B(NCE22CS101)
SNEHA SOMAKUMAR(NCE22CS107)

INTRODUCTION

- Air pollution and toxic gases pose serious risks, especially in confined spaces like wells and underground water tanks.
- Workers in such environments face dangers like oxygen deficiency and exposure to harmful gases.
- AirGuard is a smart IoT-based life-saving device for real-time air quality monitoring.
- It continuously measures carbon gases and oxygen levels to detect unsafe conditions.
- The system provides instant alerts and guidance, helping prevent accidents and fatalities.
- AirGuard is a cost-effective and reliable solution that improves safety and reduces deaths in hazardous environments.

ABSTRACT

- Air pollution and harmful gases like carbon dioxide in confined spaces (mines, tanks, factories) can be dangerous and hard to detect.

- The Smart IoT-Based Air Safety System monitors carbon dioxide and oxygen levels in real time using sensors and a microcontroller.

- The system sends data wirelessly to a platform for remote monitoring and analysis.

- If gas levels exceed safe limits or oxygen decreases, it gives alerts automatically.

- This system improves worker safety, prevents accidents, and is cost-effective and suitable for industrial and environmental use.

OBJECTIVE

- Monitor gas and oxygen levels in real time
- Detect unsafe conditions and provide instant alerts
- Improve worker safety and prevent loss of life

PROBLEM STATEMENT

- Hazardous gases and low oxygen in confined spaces like wells
- Lack of real-time monitoring and alert systems
- High risk of accidents, suffocation, and fatalities

EXISTING SYSTEM

- Most existing gas detection systems are primarily hardware-centric with limited or no software integration, reducing system flexibility and intelligence.
- These systems mainly depend on basic alert mechanisms such as buzzers and LEDs, without providing detailed or actionable information to users.
- They lack real-time monitoring and remote access, preventing users from continuously tracking gas levels from different locations.
- There is minimal or no support for data storage, which makes history tracking and long-term analysis impossible.
- Due to the absence of stored data, analytics and predictive analysis cannot be performed to foresee potential gas hazards.
- Existing systems do not support smart automation features, such as safety control based on detected gas levels.

PROPOSED SYSTEM

- A smart IoT-based emission monitoring system integrating both hardware and software layers.
- Uses MQ-135 gas sensor and AO-03 oxygen sensor with ESP-8266 to monitor air quality and harmful gas levels in real time.
- Automated safety feature: triggers a buzzer/siren when gas concentration exceeds safe limits.
- Mobile application integration: provides live data visualization, real-time alerts, and manual control of the buzzer.
- Enables wireless monitoring and control through IoT connectivity.

MODULE DESCRIPTION

SENSOR MODULE

- Consists of MQ-135, AO-03 (O_2) sensors for detecting harmful gases and oxygen levels.
- Continuously measures CO_2 concentration and oxygen percentage in the environment.
- Collects real-time air quality data without interruption.
- Sends sensor readings to the controller module for further processing.
- Forms the base of real-time environmental and gas monitoring.

CONTROLLER MODULE

- Uses ESP-8266 as the central microcontroller unit.
- Receives and processes data from all connected gas sensors.
- Applies threshold-based logic to identify unsafe gas conditions.
- Automatically activates the buzzer when CO_2 increases or O_2 decreases.
- Transmits processed data to the blynk server using Wi-Fi module.

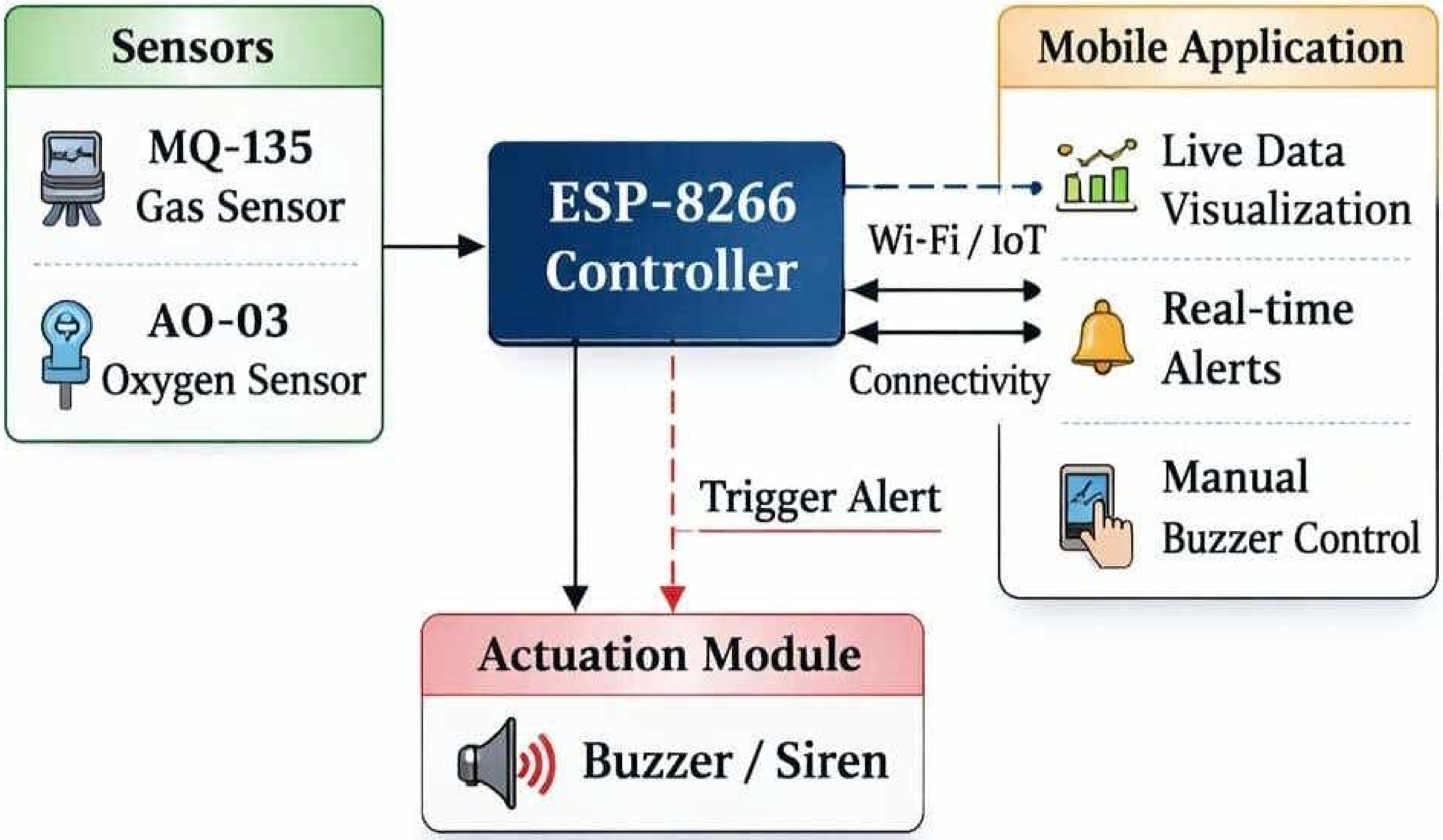
MOBILE APPLICATION MODULE

- Built using Flutter for cross-platform compatibility.
- Displays real-time gas readings and oxygen levels.
- Shows alerts and notifications during unsafe environmental conditions.
- Allows manual control of buzzer from the application.

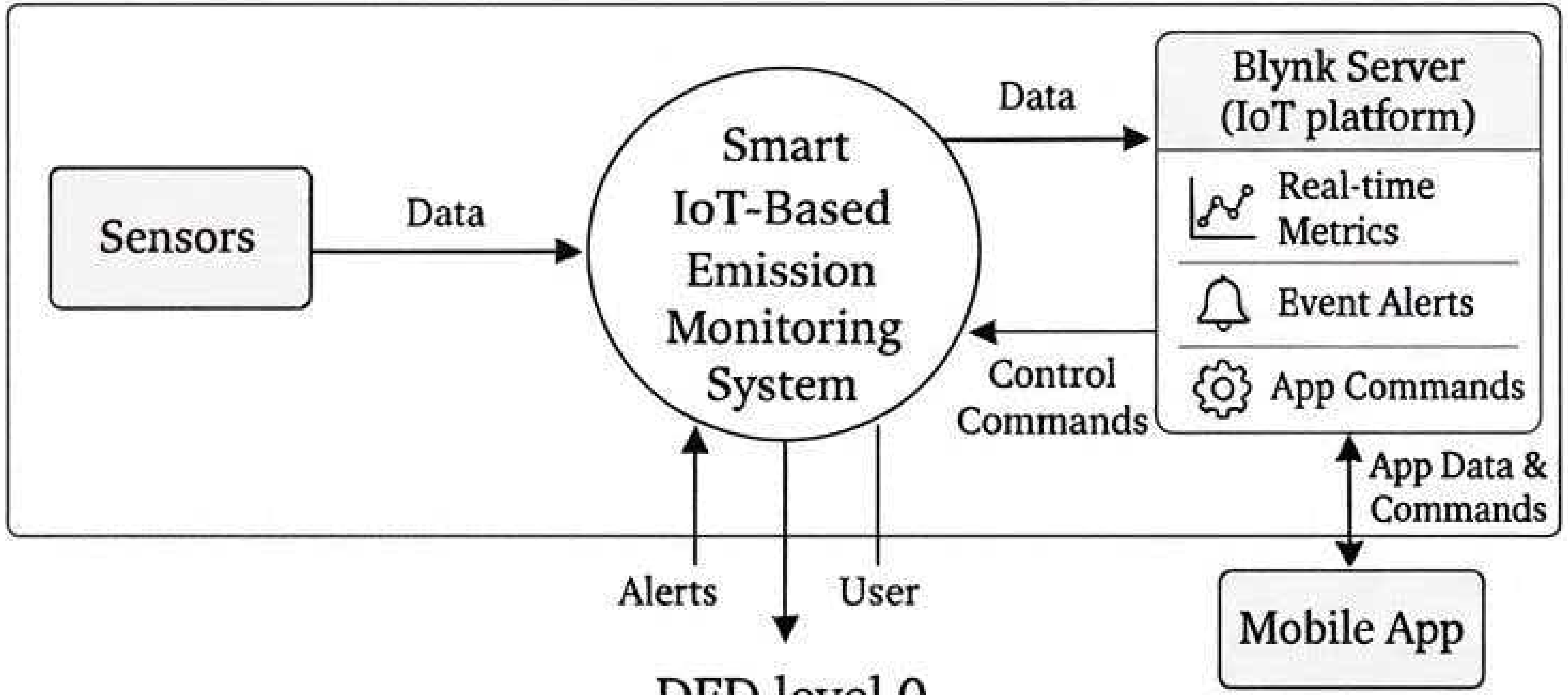
ACTUATION MODULE

- Includes a buzzer or siren to alert users during hazardous gas levels.
- Actuators are controlled by the ESP-8266 based on sensor thresholds.
- Supports manual override through the mobile application.
- Ensures immediate safety actions in critical environmental conditions.

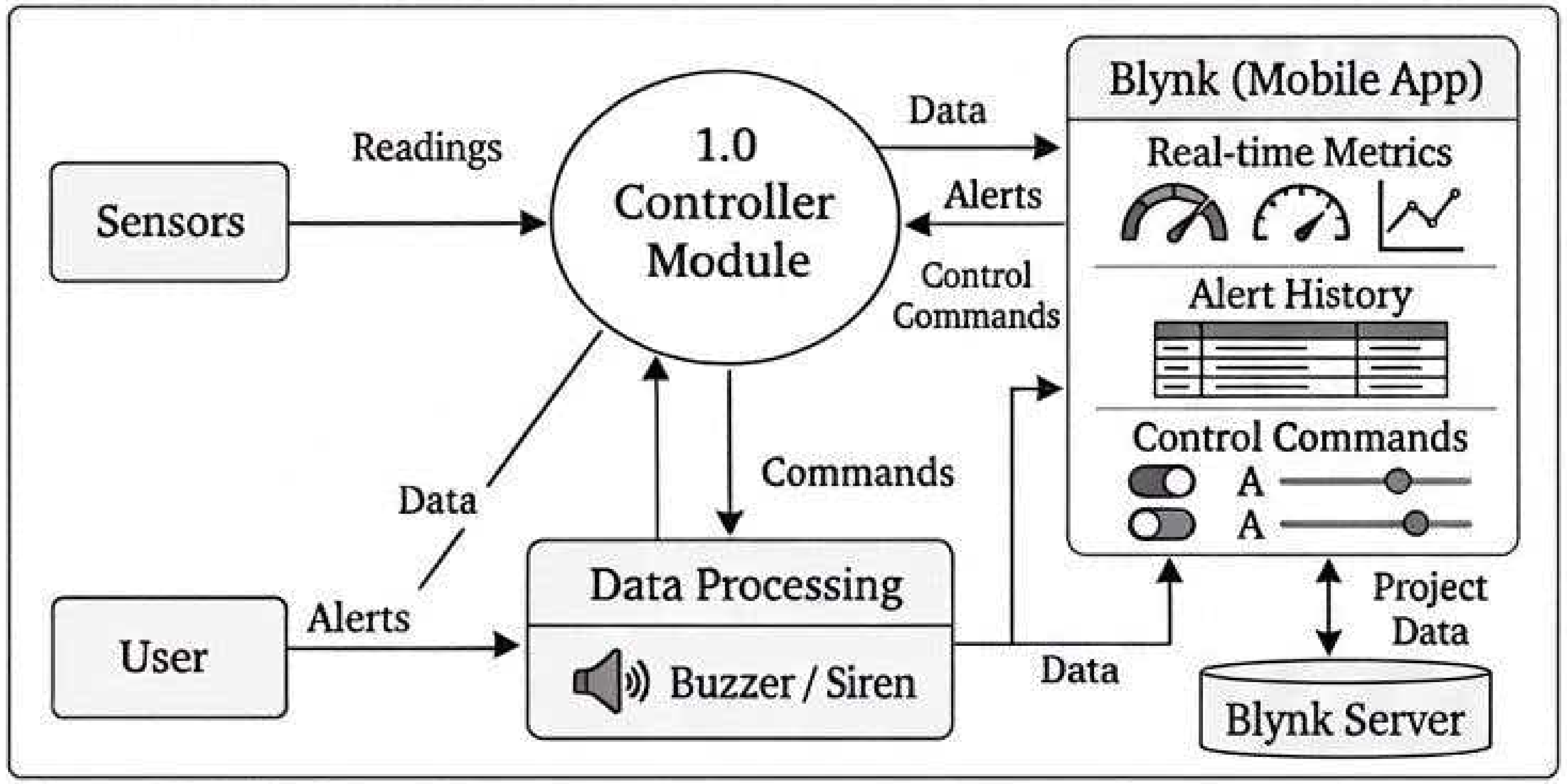
SYSTEM ARCHITECTURE



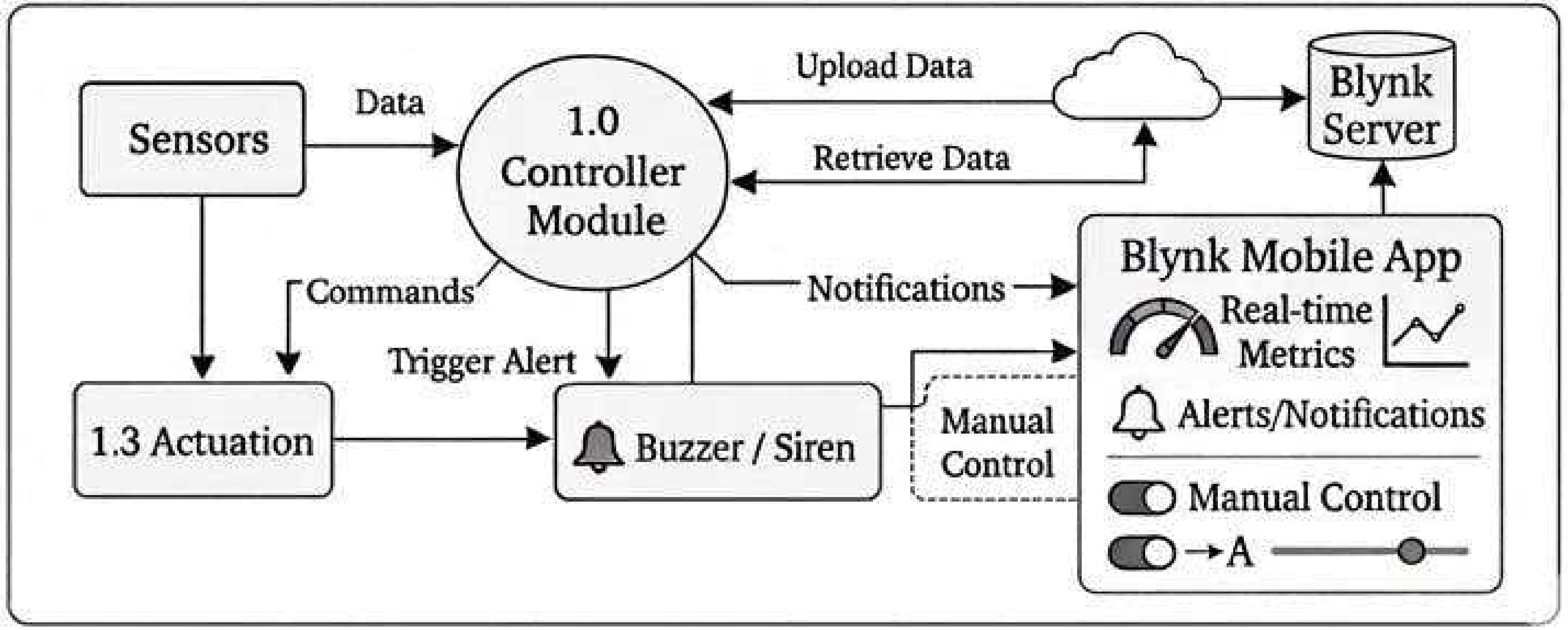
DATAFLOW DIAGRAM



DFD level 0



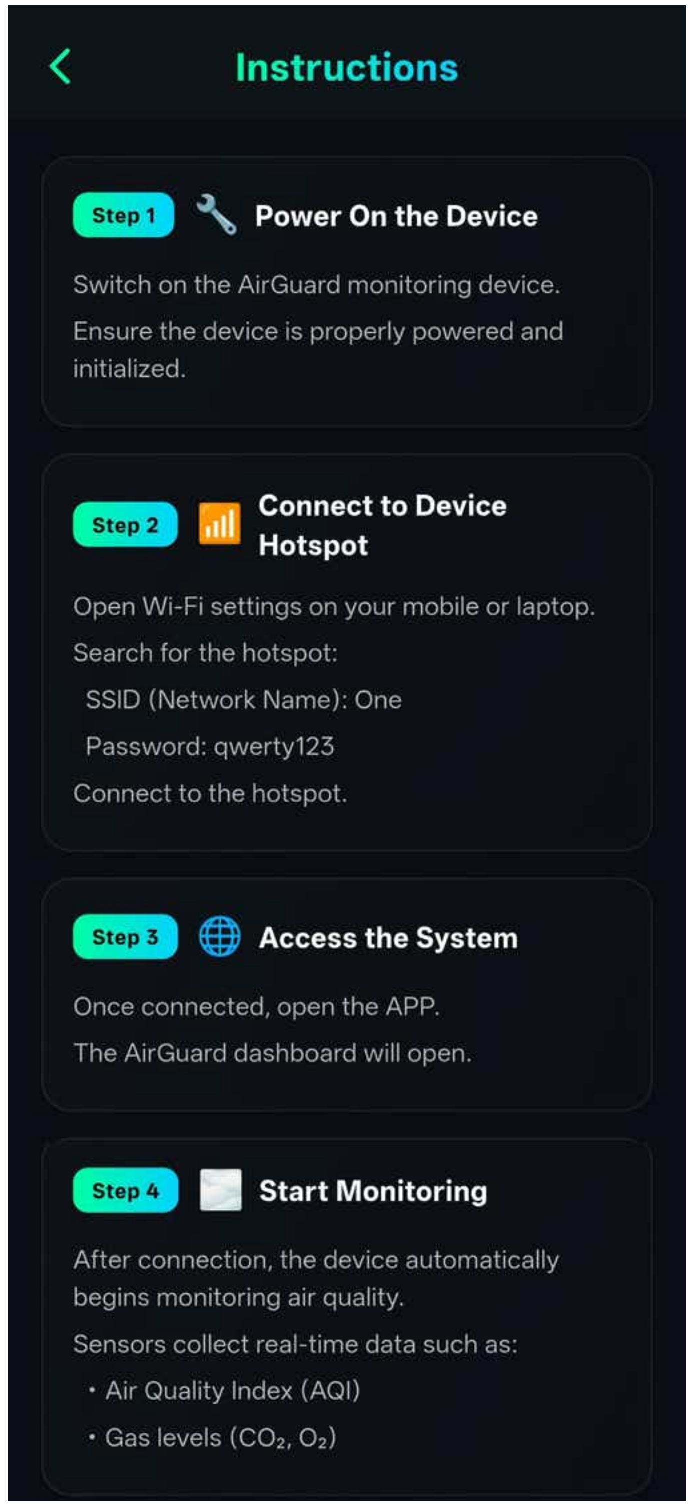
DFD level 1



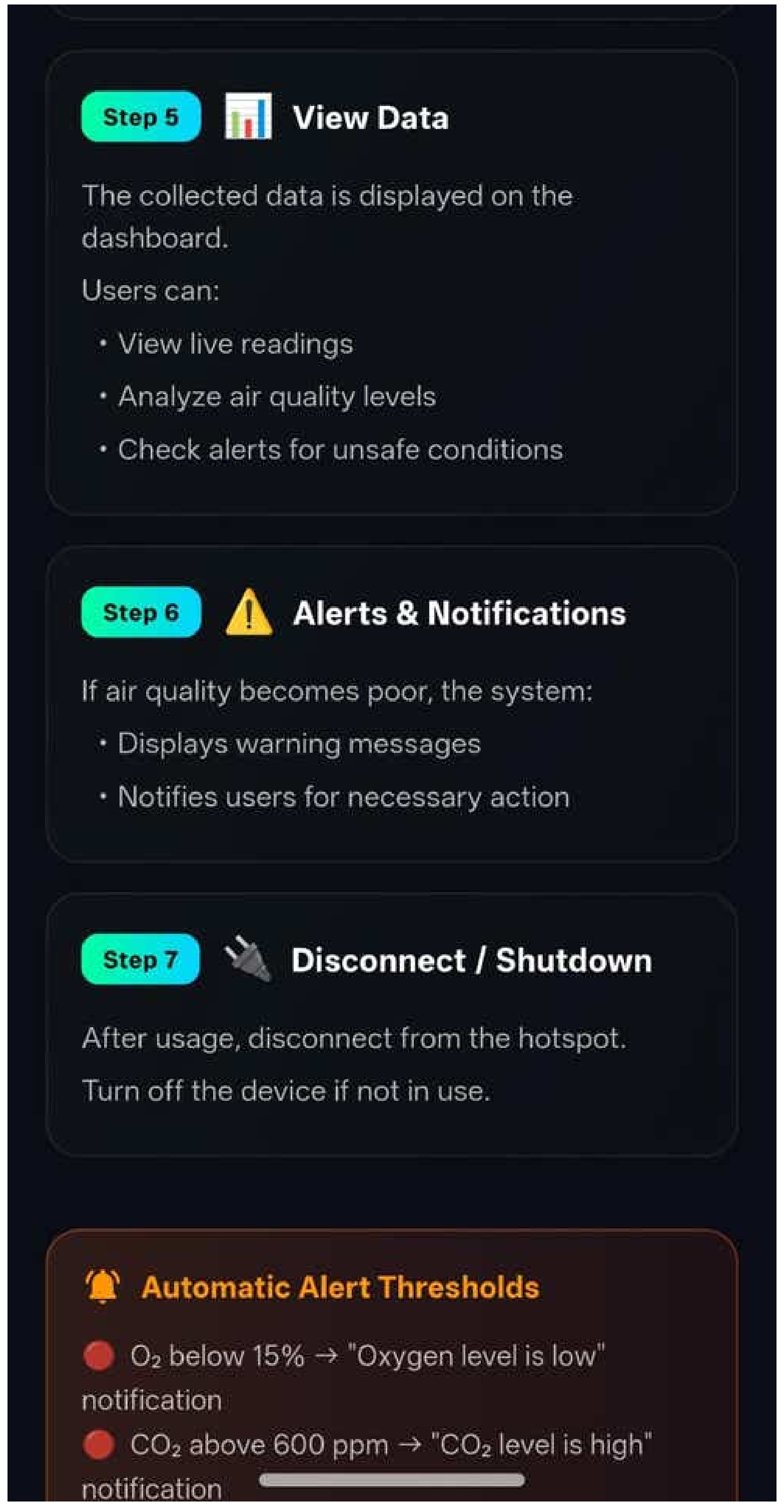
DFD level 2

INTERFACE

8:14
AirGuard Monitor
Real-time Monitoring
50
Good
Air Quality Index
Today, 24/03/2026
Live
CO_2 410 ppm
CO_2 Level
Good
20.9%
O_2 Level
Good
20.9% O_2
Oxygen Level
Air Quality Status
Oxygen vs Carbon emissions
410 ppm
CO_2 Emissions
Control Systems
Buzzer Alert System
Safety Alarm Control

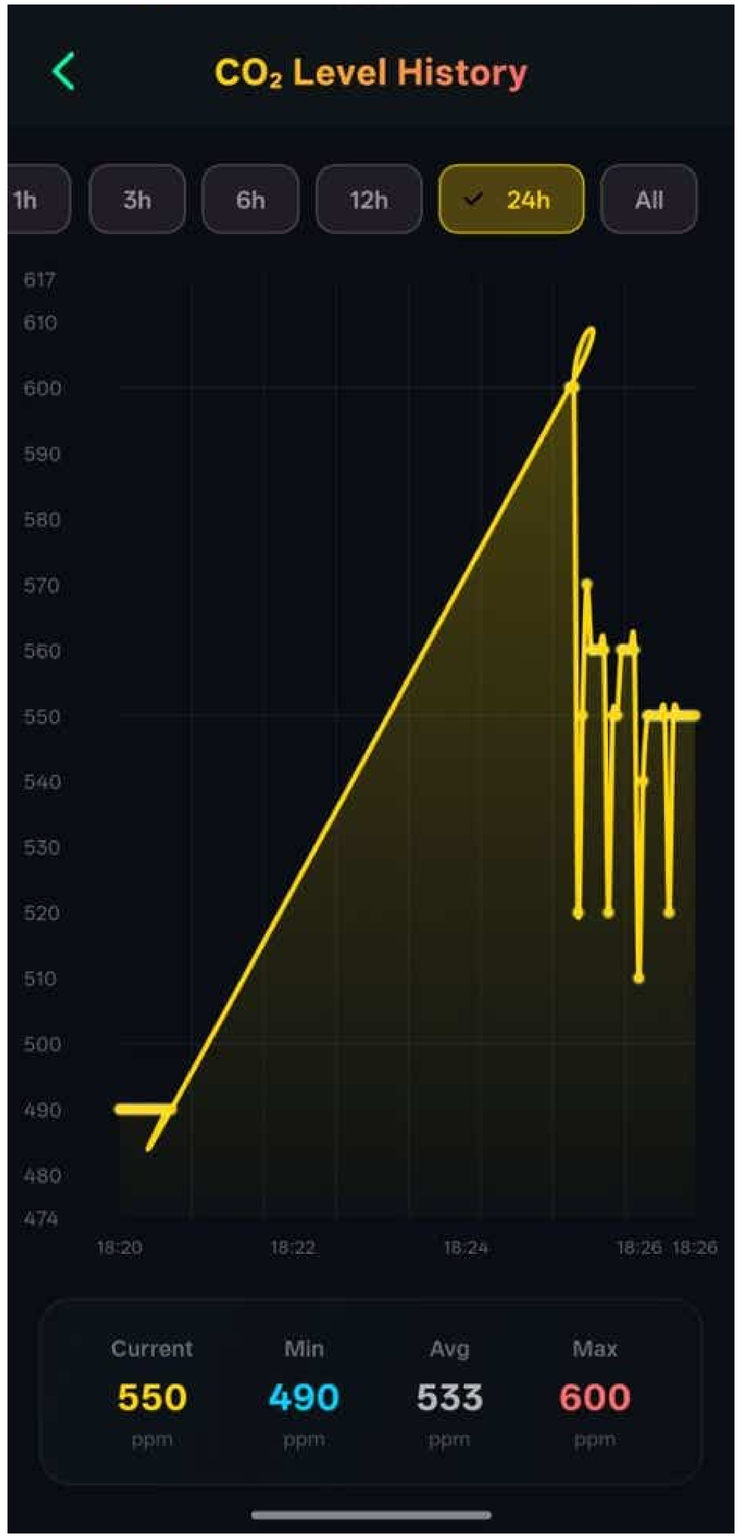


Instructions
Step 1 Power On the Device
Switch on the AirGuard monitoring device.
Ensure the device is properly powered and initialized.
Step 2 Connect to Device Hotspot
Open Wi-Fi settings on your mobile or laptop.
Search for the hotspot:
SSID (Network Name): One
Password: qwerty123
Connect to the hotspot.
Step 3 Access the System
Once connected, open the APP.
The AirGuard dashboard will open.
Step 4 Start Monitoring
After connection, the device automatically begins monitoring air quality.
Sensors collect real-time data such as:
• Air Quality Index (AQI)
• Gas levels (CO_2, O_2)



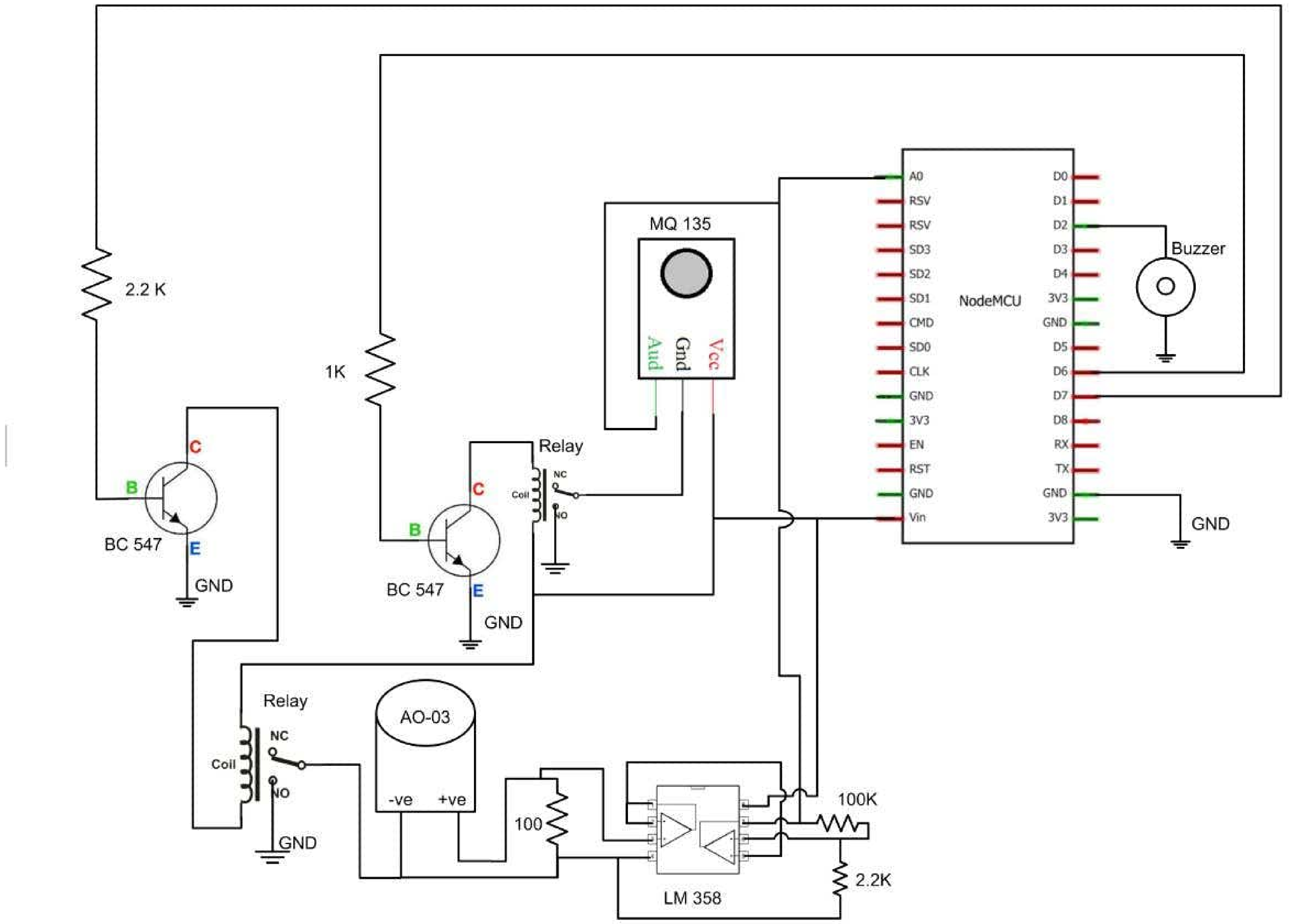
Step 5 View Data
The collected data is displayed on the dashboard.
Users can:
• View live readings
• Analyze air quality levels
• Check alerts for unsafe conditions
Step 6 Alerts & Notifications
If air quality becomes poor, the system:
• Displays warning messages
• Notifies users for necessary action
Step 7 Disconnect / Shutdown
After usage, disconnect from the hotspot.
Turn off the device if not in use.
Automatic Alert Thresholds
O_2 below 15% → "Oxygen level is low" notification
CO_2 above 600 ppm → "CO_2 level is high" notification

INTERFACE

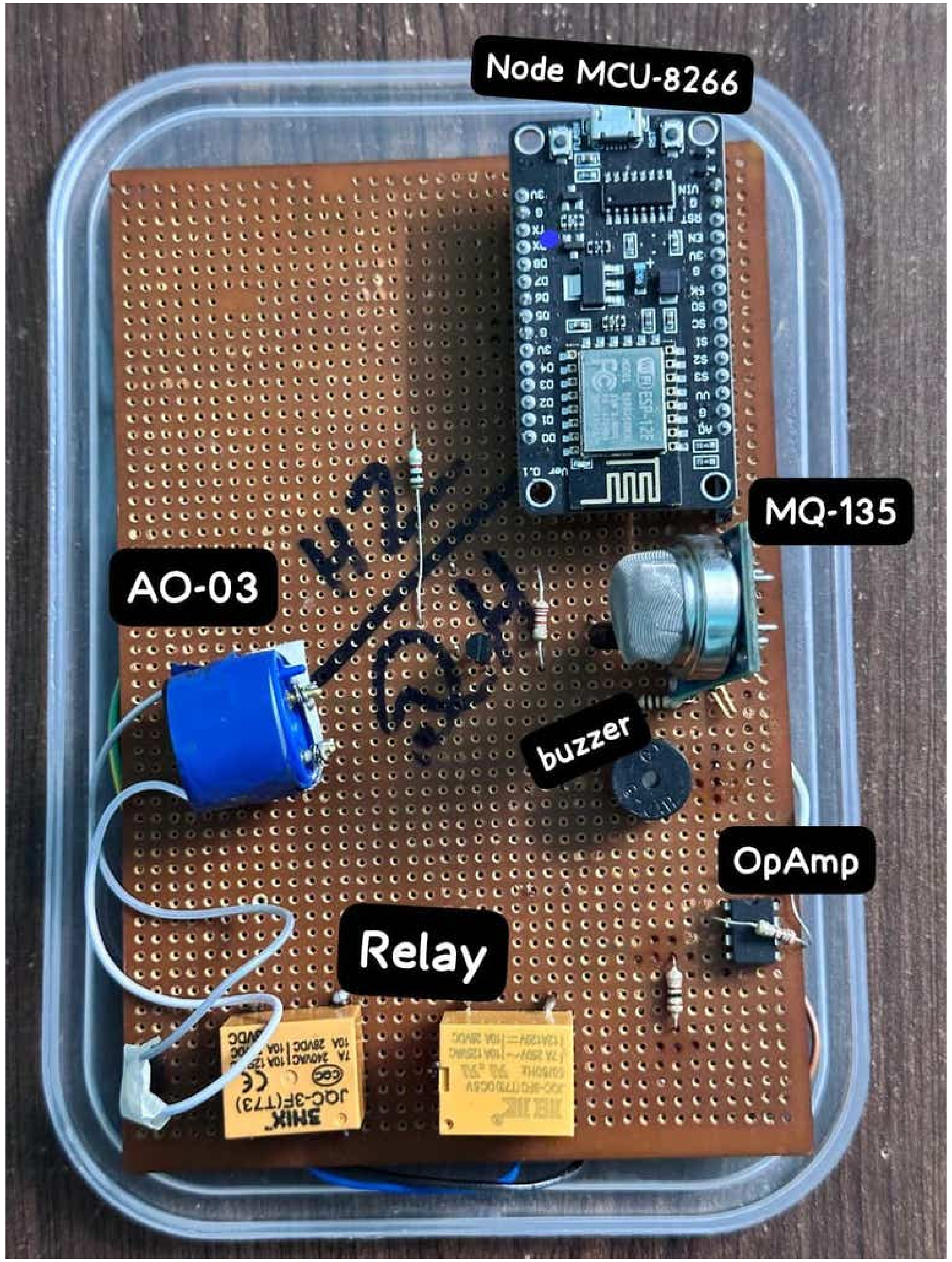

CO₂ Level History
1h
3h
6h
12h
24h
All
617
610
600
590
580
570
560
550
540
530
520
510
500
490
480
474
18:20
18:22
18:24
18:26
18:26
Current
550
ppm
Min
490
ppm
Avg
533
ppm
Max
600
ppm


O₂ Level History
1h
3h
6h
12h
24h
All
25.9
25.0
24.0
23.0
22.0
21.0
20.0
19.0
18.0
17.0
16.0
15.6
18:20
18:22
18:24
18:26
18:27
Current
20.9
%
Min
20.6
%
Avg
20.9
%
Max
20.9
%

CIRCUIT DIAGRAM



HARDWARE



COMPONENTS USED:

1. Relay :

- Switches between Oxygen and CO_2 sensors
- Controlled by microcontroller
- Ensures safe sensor selection

2. Operational Amplifier :

- Amplifies weak sensor signals
- Suitable for ADC interfacing
- Low power consumption

3. ESP8266 :

- Provides Wi-Fi connectivity
- Sends sensor data to cloud
- Low power and compact

4.MQ-135 (CO_2 Sensor):

- Detects CO_2 concentration
- Provides analog output
- Used for air quality monitoring

5.AO-03 (Oxygen Sensor):

- Measures oxygen level
- Provides analog output
- High sensitivity

FUTURE SCOPE

- AI-Based Prediction:Use Machine Learning to forecast hazardous gas conditions in advance.
- Advanced Data Analytics:Generate detailed insights, reports, and visual trends from collected data.
- Multi-Gas Detection:Extend the system to detect gases like CO, LPG, and methane.
- Smart Automation:Automatically control ventilation and safety systems during danger.
- Smart City Deployment:Implement in public areas for large-scale environmental monitoring.
- Portable & Wearable Devices:Develop compact versions for workers in confined and hazardous environments.

CONCLUSION

- The proposed system provides real-time monitoring of CO_2 and O_2 levels to ensure a safe environment.
- Integration of sensors, controller and mobile application enables smart and automated gas detection.
- Automatic and manual actuation ensures immediate response during hazardous conditions.
- The mobile application improves user awareness through real-time alerts and visual indicators.
- Overall, the system enhances safety, reliability, and scalability compared to traditional gas detection systems.

REFERENCES

[1] Atzori, L., Iera, A., & Morabito, G. “The Internet of Things: A Survey,” Comput. Netw., vol. 54, no. 15, pp. 2787–2805, October 2010. [7]

[2] Bandyopadhyay, D., & Sen, J. “Internet of Things: Applications and Challenges in Technology and Standardization,” Wirel. Pers. Commun., vol. 58, no. 1, pp. 49–69, May 2011. [6]

[3] Devarakonda, S., Sevusu, P., Liu, H. H., Liu, R., Iftode, L., & Nath, B. “Real-Time Air Quality Monitoring Through Mobile Sensing In Metropolitan Areas,” ACM Urbcomp, August 2013. [12]

[4] Gluhak, A., Krco, S., Nati, M., Pfisterer, D., Mitton, N., & Razafindralambo, T. “A Survey on Facilities for Experimental Internet of Things Research,” IEEE Communications Magazine, vol. 49, no. 11, pp. 58–67, November 2011. [9]

Thank You